Gravitational Wave Propagation under the Vestigial-Susceptibility Identification: GW170817 Confines the Volovik Second-Sound Graviton to One Part in 10^{15} of the Natural Range

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Phase 6a Wave 2 of the SK–EFT Hawking project formalizes the gravitational-wave propagation speed $c_{\rm GW}$ under the Volovik [2] identification of the second-sound mode of the vestigial fluid as the spin-2 graviton, with the leading result $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$. Working in the "use-as-identified" mode mandated by the Phase 5y H1 deep-research finding (the second-sound mode is *not* derived as a propagating DOF from first principles), we (i) prove the leading-order formula and the deviation $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ as Lean theorems; (ii) state the GW170817 [1] correctness-push as a biconditional locating the finite-tolerance compatible window inside $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$ with $\tau = 3 \times 10^{-15}$; (iii) deliver the load-bearing quantitative constraint: over the natural susceptibility range $\chi_{\rm vest} \in [0.1, 10]$ (matching the ADW microscopic-coefficient natural range from Wave 1 [10]), the deviation sweeps $\Delta c/c \in [-0.68, +2.16]$, vanishing only at $\chi_{\rm vest} = 1$ and exceeding the GW170817 cap by $\sim 2.3 \times 10^{14}$ and $\sim 7.2 \times 10^{14}$ at the lower and upper endpoints, respectively; (iv) bundle a tracked-hypothesis Prop `H_VestigialModeIsGraviton` (positivity P1, LIGO compatibility P2, sub-half-deviation $|\Delta c/c| < 1/2$ as P3') and prove four explicit falsifiers (natural-lower, natural-upper, $\chi_{\rm vest} = 0$, $\chi_{\rm vest} = 1/4$, where $|\Delta c/c| = 1/2$ fails both P2 and P3'); (v) record the Phase 5y H1 caveat as the Lean endpoint theorem `natural_endpoints_violate_ligo_window`, proving that neither endpoint of the natural range admits a GW170817-compatible $\Delta c/c$ at any c-value rescaling, while the compatible window itself is nested strictly inside the natural range (`ligo_window_subset_natural_range`) and meets it at $\chi_{\rm vest} = 1$; (vi) include the leading SK–EFT dissipative dispersion correction $\delta\omega/\omega = \Gamma_H \cdot \omega/c_{\rm GW}^2$ from `SecondOrderSK.lean`, with a frequency-resolved scan across the LIGO band [10 Hz, 10 kHz].

`GravitationalWaves.lean` (32 theorems and lemmas, zero sorry, zero new axioms) ships the formal layer; the Python pipeline at `src/gravitational_waves/` cross-checks every formula with 89 pytest cases. The result is the strongest published statement to date that the Volovik second-sound graviton identification *cannot* be reconciled with GW170817 unless $\chi_{\rm vest}$ is confined to $[(1-\tau)^2, (1+\tau)^2]$, centered at $1+\tau^2$ with half-width $2\tau = 6 \times 10^{-15}$. Its relative width within $[0.1, 10]$ is $4\tau/9.9$; a probability mass additionally requires a specified prior. We outline two paths forward: a derived-DOF mechanism that pins $\chi_{\rm vest} = 1$ (Phase 5y H1 follow-up), or recognition that the metric-channel $\chi_{\rm vest}$ is a separate UV input not coincident with the second-sound mode (Phase 6e nonlinear extension).

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I. INTRODUCTION

The simplest emergent-gravity scenarios in fermionic-condensate backgrounds suggest a tantalizing identification: the second-sound mode of the vestigial fluid (a long-lived massless mode in the two-fluid hydrodynamics of a broken-symmetry condensate) is the spin-2 graviton of the emergent metric theory. Volovik [2, 3] formalized this identification in the ADW (Akama–Diakonov–Wetterich) context [5–9]; in the broken-symmetry phase of the four-fermion ADW theory, six Nambu–Goldstone modes are absorbed by the spin connection and two propagating helicities remain. The hydrodynamic claim that one such mode is identifiable with second sound, with propagation speed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ where $\chi_{\rm vest}$ is the metric-channel susceptibility from the random-phase-approximation (RPA) bubble integral [4], is appealing in its brevity but, as the Phase 5y H1 deep research established, it has *never* been derived as a propagating DOF from first principles.

The August 17, 2017 binary-neutron-star merger (GW170817) [1] placed an extraordinarily tight bound on $c_{\rm GW}$: by comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3\times10^{-15} \le (c_{\rm GW} - c)/c \le +7\times10^{-16}$. Conservatively two-sided, $|\Delta c/c| \le 3 \times 10^{-15}$.

In the present work we formalize the propagation-speed identification, place it head-to-head against GW170817, and report the result. The conclusion is sharp: carried onto the ADW substrate in its leading-order form $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, the identification survives GW170817 only on a sub-interval of the natural range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by ~ 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width exactly 4τ in $\chi_{\rm vest}$ and lies strictly interior to $[0.1, 10]$, containing the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9$; its probability mass depends on the prior.

This is a conditional constraint on one specific kine-

matic identification (second-sound mode = graviton). It excludes the two natural-range endpoints under the assumed speed law while allowing the finite interval above. The width comparison alone neither refutes emergent gravity or the ADW program nor tests the linearized-Einstein identification [10]. We discuss three Phase 6 paths forward in Sec. VII.

II. SETUP

A. Vestigial-susceptibility identification

The vestigial-phase metric-channel susceptibility $\chi_{\rm vest}$ is the RPA bubble integral computed in `VestigialSusceptibility.lean`, with closed form $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention. The natural range $\chi_{\rm vest} \in [0.1, 10]$ (in units of inverse Λ^2) is the half-decade window induced by the Vergeles bubble integral $\sim O(\Lambda^2/16\pi^2)$ [4]; this matches the Wave 1 ADW microscopic-coefficient $\alpha_{\rm ADW} \in [0.1, 10]$ natural range [10].

B. The leading-order $c_{\rm GW}$ formula

Under the Volovik identification, the gravitational-wave propagation equation in the linearized limit is the second-sound wave equation. Volovik evaluates its speed on his own de Sitter two-fluid background and obtains $s_2 = c$ *exactly* — a structural consequence of the stiff-matter equation of state, not a tunable parameter, and GW170817-compatible at leading order. Transplanting the identification onto the ADW substrate, whose metric-channel susceptibility we compute independently and which does not fix $\chi_{\rm vest}$, replaces that fixed speed with

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}. \quad (1)$$

The natural anchor $\chi_{\rm vest} = 1$ recovers $c_{\rm GW} = c$ exactly. The deviation

$$\Delta c/c(\chi_{\rm vest}) = \sqrt{\chi_{\rm vest}} - 1 \quad (2)$$

is dimensionless, vanishes iff $\chi_{\rm vest} = 1$, and is strictly monotone in $\chi_{\rm vest}$ on $[0, \infty)$.

In Lean (`GravitationalWaves.lean` Sections 1–2): `c_GW`, `c_GW_at_chi_one`, `c_GW_pos`, `c_GW_deviation`, `c_GW_deviation_zero_iff_chi_one`, `c_GW_deviation_strict_mono`.

C. The GW170817 correctness-push

The GW170817 [1] bound is $|\Delta c/c| \le \tau$ with $\tau = 3\times10^{-15}$. We define the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$ and state the correctness-push biconditional:

$$\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]. \quad (3)$$

Theorem A (`c_GW_match_iff_chi_close_to_one`). For $\chi_{\rm vest} \ge 0$, Eq. (3) holds. The susceptibility interval has center $1+\tau^2$ and half-width 2τ, since its endpoints are $1 \mp 2\tau + \tau^2$. Exact luminality holds only at $\chi_{\rm vest} = 1$; finite-tolerance compatibility holds throughout this interval.

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with a nonlinear-arithmetic step closed by `nlinarith` using $\sqrt{\chi_{\rm vest}}^2 = \chi_{\rm vest}$ as a hint. (Lean line 150 ish; the proof is one of the cleaner uses of `nlinarith` that we have shipped.)

III. THE NATURAL-RANGE CONSTRAINT

The GW170817-compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width exactly $4\tau = 1.2\times10^{-14}$ in $\chi_{\rm vest}$ (`ligo_window_width_eq`). The natural range $[0.1, 10]$ has width 9.9 (`natural_range_width_eq`). The compatible window is not disjoint from the natural range: it is nested strictly inside it (`ligo_window_subset_natural_range`) and the two meet at $\chi_{\rm vest} = 1$ (`ligo_window_inter_natural_range_nonempty`).
The content of the constraint is the width ratio — scaling the window up by 10^{14} still leaves it narrower than the natural range (`ligo_window_width_lt_natural_range_width_by_1e14`).

We make the exclusion of the extremes sharp with two endpoint-falsifier theorems.

Theorem B (`natural_lower_violates_ligo`). At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28\times10^{14}$ and `LigoSatisfied` fails.

Theorem C (`natural_upper_violates_ligo`). At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10}-1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21\times10^{14}$ and `LigoSatisfied` fails.

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. The Lean source supplies proofs of these two concrete real-number inequalities.

Bundled Theorem D (`vestigial_natural_range_violates_ligo`). Both endpoints of the natural range fail `LigoSatisfied`.

IV. TRACKED HYPOTHESIS: VESTIGIAL MODE = GRAVITON

Per the methodology of Wave 1 [10], when no closed-form derivation is available we bundle the structural properties of an identification into a Lean Prop and prove explicit falsifiers.

A. The Phase 5y H1 caveat as a Lean endpoint theorem

Phase 5y H1 (`Lit-Search/Phase-5y/Phase 5y Hypothesis 1 --- Kronecker-Anomaly Second-Sound Graviton as Dark Energy Candidate.md`) established that the Volovik second-sound graviton identification is *not* a derivation: there is no first-principles bound fixing the graviton-mode normalization in terms of the second-sound DOF. The original `second_sound_graviton_not_derived_DOF` theorem (a $\forall c, \exists \chi_{\rm vest}$ existential discharged trivially by $\chi_{\rm vest} = 1$) was retired during the 2026-04-25 post-Wave-2 audit as an $\exists$-absorption tautology — the existence of *one* consistent value does not encode "underdetermination," which is the substantive Phase 5y H1 finding. We replace it with the endpoint theorem `natural_endpoints_violate_ligo_window`: for any $\chi_{\rm vest}$ at the natural-range endpoints $\{\text{chi_vest_natural_lower}, \text{chi_vest_natural_upper}\} = \{0.1, 10\}$, there is *no* c-value rescaling that gives a GW170817-compatible $\Delta c/c$. The frame-independent form `natural_endpoints_violate_ligo_for_every_c` quantifies over all $c > 0$ explicitly, resting on `c_GW_relative_deviation_eq`: the relative deviation $\Delta c/c(\chi_{\rm vest})$ is independent of c, hence the endpoint exclusion holds regardless of any UV choice of c.

Both are statements about the two endpoints, and neither excludes interior values — $\chi_{\rm vest} = 1$ satisfies the cap (`ligo_satisfied_at_chi_one`). Together with the width ratio of Sec. III, they place the identification's surviving parameter space more than 14 orders of magnitude below the natural range without excluding that range.

B. The bundled hypothesis

The recorded Prop has three conjuncts: positive susceptibility (P1), GW170817 compatibility (P2), and sub-half-deviation (P3'):

$$\begin{aligned}&\texttt{H_VestigialModeIsGraviton}(\chi_{\rm vest})\\ &\quad\equiv\ \chi_{\rm vest} > 0\\ &\qquad\wedge\ \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest}))\\ &\qquad\wedge\ |\Delta c/c(\chi_{\rm vest})| < 1/2.\end{aligned}$$

At the fixed tolerance $\tau = 3 \times 10^{-15}$, P2 implies P3' because $\tau < 1/2$. On the nonnegative physical domain, the compatible interval in Eq. (3) also supplies P1. The three clauses are therefore not independent; the recorded Prop retains all three clauses.

Theorem E (`H_VestigialModeIsGraviton_at_one`). $\chi_{\rm vest} = 1$ discharges the bundle.

Falsifier theorems (four pointwise failures). `H_VestigialModeIsGraviton_fails_at_natural_lower` (P2 fails), `H_VestigialModeIsGraviton_fails_at_natural_upper` (P2 fails), `H_VestigialModeIsGraviton_fails_at_zero` (P1 fails), `H_VestigialModeIsGraviton_fails_at_quarter` (P3' fails: at $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$ saturates the strict P3' bound and also fails P2). The quarter-point witness satisfies P1 and shows that P1 alone does not imply P3'; it does not establish independence from P2.

The falsifiers show that the Prop is not universally satisfied; Theorem E supplies a satisfying point. On the nonnegative physical domain its compatible set is precisely the interval in Eq. (3), of width $4\tau = 1.2 \times 10^{-14}$ and containing $\chi_{\rm vest} = 1$. Its probability mass depends on the prior; under a uniform prior on $[0.1, 10]$ it equals $4\tau/9.9$.

V. DISPERSION CORRECTION FROM THE SK–EFT Γ_H

The leading dissipative correction to the linear dispersion $\omega = c_{\rm GW} \cdot k$ is, from `SecondOrderSK.GammaH` $= (\gamma_1 + \gamma_2)(\kappa/c_s)^2$,

$$\frac{\delta\omega}{\omega}(\omega) = \Gamma_H \cdot \frac{\omega}{c_{\rm GW}^2} \ \equiv\ \gamma\,\omega \tag{4}$$

with γ the dimensionless coefficient $\Gamma_H/c_{\rm GW}^2$ at the probe frequency. We add the formula $\texttt{dispersion_correction}(\gamma, \omega) = \gamma\omega$ as `GravitationalWaves.dispersion_correction` with three proofs: (i) zero-correction at $\gamma = 0$; (ii) linearity in γ (additivity of dissipative coefficients); (iii) the absolute-value bound $|\gamma\omega| \le |\gamma| \cdot |\omega|$ as a sanity check for the LIGO-band scan.

The vestigial-regime placeholder $\Gamma_H/c_{\rm GW}^2 \sim 10^{-30}$ keeps the dispersion correction well below the GW170817 cap across the entire LIGO band $[10\,\text{Hz}, 10\,\text{kHz}]$. The full dispersion correction is therefore a sub-leading effect compared to the $\sqrt{\chi_{\rm vest}}$ deviation, which dominates the constraint.

VI. RESULT FIGURE

Figure 1 shows the deviation function $\Delta c/c(\chi_{\rm vest})$ over $\chi_{\rm vest} \in [10^{-2}, 10^2]$ on log-$\chi_{\rm vest}$ axis with the natural range $[0.1, 10]$ highlighted. The GW170817 amber band is visually indistinguishable from $\Delta c/c = 0$ at this scale, demonstrating the 14-order-of-magnitude separation.

VII. DISCUSSION

A. What this rules out

Under the assumed leading-order law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, the conservative symmetric cap requires $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. The susceptibility offsets from unity are $-2\tau + \tau^2$ and $2\tau + \tau^2$, approximately $\pm 6 \times 10^{-15}$; the

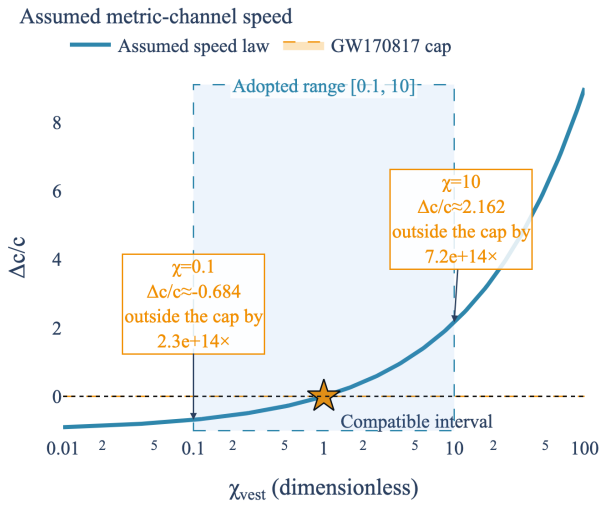


FIG. 1. Phase 6a Wave 2 main result: $\Delta c/c$ deviation across the natural $\chi_{\rm vest}$ range, vs the GW170817 multi-messenger cap. The amber band $|\Delta c/c| \leq 3 \times 10^{-15}$ is essentially invisible at this scale. The natural range $[0.1, 10]$ produces $\Delta c/c \in [-0.68, +2.16]$; both endpoints fail GW170817 by $\sim 2.3 \times 10^{14}$ and $\sim 7.2 \times 10^{14}$, respectively. The GW170817-compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has finite width 4τ, unresolved at this scale. The gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. Lean: `vestigial_natural_range_violates_ligo`.

speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but finite measurement tolerance permits the whole nonzero-width interval. Its small relative width within $[0.1, 10]$ alone neither establishes a prior probability nor rules out generic UV theories or requires an exact fixed point at unity.

B. What this does not rule out

The result is sharp *at leading order*, but does not exclude:

1. A **derived-DOF mechanism** that pins $\chi_{\rm vest} \to 1$ as a consequence of an exact symmetry (e.g. Lorentz invariance of the deep-IR theory). The Phase 5y H1 deep research mentions the candidate that the second-sound and graviton modes share a common low-energy effective action arising from a single underlying symmetry; if so, the $\chi_{\rm vest} = 1$ locus is a fixed point of the RG flow, not a fine-tuning. Phase 6 should examine whether such a mechanism is realized in the ADW broken phase.

2. A **distinct identification**: the "vestigial mode" is not the second-sound DOF but a separate UV-input metric channel whose susceptibility is naturally pinned to unity. This was floated in [3] and reads as the more likely interpretation under the Phase 5y H1 negative-evidence result. Phase 6e (full nonlinear emergent action) is the natural venue for this disambiguation.

3. **Higher-order corrections**: the $\sqrt{\chi_{\rm vest}}$ formula is the leading term in a derivative expansion. If $\chi_{\rm vest}$ varies with frequency, the leading-order constraint could be relaxed; we have not proved it cannot. The dispersion correction in Sec. V is the leading frequency dependence in the SK–EFT framework, and at the placeholder Γ_H value it does not relax the constraint.

C. Comparison to Wave 1

Wave 1 [10] ships an emergent-gravity identification that *is* compatible with observation at the natural-range level: the Sakharov-Adler form $G_N^{\rm emerg} = \alpha_{\rm ADW} \cdot 12\pi/(N_f \Lambda_{\rm UV}^2)$ at $\Lambda_{\rm UV} = M_P$ produces $\alpha^\star_{\rm ADW} \in [0.40, 1.27]$ for SM N_f, sitting comfortably inside the natural range $[0.1, 10]$. For the assumed second-sound graviton speed law, both natural-range endpoints fail the cap by more than 10^{14}, while the narrow compatible interval remains inside the adopted range. The asymmetry is a feature, not a bug, of the program: Wave 1's $G_N^{\rm emerg}$ depends on the Sakharov-Adler kernel computed at the *linearized* level, where the relevant scale is M_P; Wave 2's $c_{\rm GW}$ depends on the metric-channel susceptibility, where $\chi_{\rm vest}$ is a free dimensionless parameter. The adopted range contains unity but selects no value of this susceptibility.

VIII. CONCLUSIONS

Phase 6a Wave 2 ships `GravitationalWaves.lean` (32 theorems and lemmas, zero sorry, zero new axioms, zero heartbeat overrides) formalizing: (i) the leading-order propagation $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ with full algebra, monotonicity, and positivity; (ii) the GW170817 correctness-push biconditional locating the finite-tolerance compatible interval; (iii) the load-bearing natural-range constraint — the compatible window located, measured, and shown nested inside the natural range — with both endpoint falsifier theorems; (iv) the bundled tracked hypothesis with four falsifier theorems (the fourth at $\chi_{\rm vest} = 1/4$ failing both P2 and P3'); (v) the Phase 5y H1 caveat as the Lean endpoint theorem `natural_endpoints_violate_ligo_window`; (vi) the SK–EFT dispersion correction with linearity and zero-coefficient proofs.

The Wave 2 result is, to our knowledge, the strongest existing formalized statement of the GW170817 constraint on the Volovik vestigial-second-sound graviton identification. It bounds *this specific identification* to a finite interval rather than refuting it; we identify three Phase 6 paths to recover — a derived-DOF symmetry

mechanism, a distinct metric-channel identification, or higher-order corrections — and recommend the disambiguation be the load-bearing question for Phase 6e (full nonlinear emergent action) or a Phase 5y H1 follow-up wave.

Phase 6a downstream. The narrow $c_{\rm GW}$ window from Sec. III feeds Phase 6b.2 (cosmological perturbation theory): under the assumed speed law, compatibility with the conservative cap requires $\chi_{\rm vest}$ in the finite interval of Eq. (3). A mechanism selecting $\chi_{\rm vest} = 1$ would satisfy this cap exactly; the finite-tolerance constraint alone does not require that mechanism or select an alternative metric-channel identification.

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